THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS · NCERT-TYPE UNIT TEST SET 3

Chapters 1–4: **Coordinates, Linear Polynomials, World of Numbers, Algebraic Identities** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The number of decimal places in the expansion of $\frac{7}{2^2 \times 5^4}$ is: 1

(a) 6 **(b)** 3 **(c)** 5 **(d)** 4

2. $(a+b)^3$ equals: 1

(a) a^3+b^3 **(b)** $a^3+3ab+b^3$ **(c)** $a^3+3a^2b-3ab^2+b^3$ **(d)** $a^3+3a^2b+3ab^2+b^3$

3. The distance between $(-1,-1)$ and $(4,11)$ is: 1

(a) 13 **(b)** 14 **(c)** 15 **(d)** 12

4. Which of these is a linear polynomial? 1

(a) $5-2x$ **(b)** x^3 **(c)** x^2-5 **(d)** 9

5. $x = 0.\overline{45}$ equals: 1

(a) $\frac{6}{11}$ **(b)** $\frac{5}{11}$ **(c)** $\frac{5}{12}$ **(d)** $\frac{4}{11}$

6. $(3x+4y)^2 =$ 1

(a) $9x^2-24xy+16y^2$ **(b)** $9x^2+16y^2$ **(c)** $9x^2+12xy+16y^2$ **(d)** $9x^2+24xy+16y^2$

7. The midpoint of the segment joining $(-6,-2)$ and $(2,10)$ is: 1

(a) $(2,4)$ **(b)** $(-2,-4)$ **(c)** $(-2,4)$ **(d)** $(4,-2)$

8. If $x=4$ is a zero of the polynomial $3x+k$, then $k=$ 1

(a) -13 **(b)** -10 **(c)** -12 **(d)** -11

9. An irrational number between 3 and 4 is: 1

(a) $\sqrt{18}$ **(b)** $\sqrt{17}$ **(c)** $\sqrt{7}$ **(d)** $\sqrt{11}$

10. $(2x-5y)^2 =$ 1

(a) $4x^2-10xy+25y^2$ **(b)** $4x^2+20xy+25y^2$ **(c)** $4x^2+25y^2$ **(d)** $4x^2-20xy+25y^2$

11. A point on the x-axis at distance 4 units to the left of the origin has coordinates: 1

(a) $(4,0)$ **(b)** $(-3,0)$ **(c)** $(-4,0)$ **(d)** $(0,-4)$

12. The value of $2x+7$ at $x=-4$ is: 1

(a) -1 **(b)** -2 **(c)** 0 **(d)** 1

13. The product of a non-zero rational number and an irrational number is: 1

(a) rational **(b)** zero **(c)** an integer **(d)** irrational

14. $(x+8)(x-5) =$ 1

(a) $x^2+3x-40$ **(b)** $x^2-3x-40$ **(c)** $x^2+13x-40$ **(d)** $x^2+3x+40$

15. If $x>0$ and $y<0$, the point (x, y) lies in: 1

(a) Quadrant I **(b)** Quadrant II **(c)** Quadrant IV **(d)** Quadrant III

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Find a rational number between 3 and 4. 2

17. Expand $(x+5)(x+8)$. 2

18. Find the distance between $P(-4, 3)$ and $Q(4, -12)$. 2

19. Find the zero of $5x+20$ and show that it lies on the x-axis of the graph $y=5x+20$. 2

20. Express $0.\overline{6}$ as a fraction in lowest terms. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Express $0.\overline{15}$ in the form $\frac{p}{q}$. 3

22. Expand $(3x-4y)^2$ and verify for $x=1$, $y=1$. 3

23. Find the point on the x-axis which is equidistant from $A(-5, 2)$ and $B(3, -2)$. 3

24. The line $y=3x+k$ has slope 3 and passes through $(2, 4)$. Find k and write the line. 3

OR

Express $0.2\overline{7}$ in the form $\frac{p}{q}$. 3

25. Factorise $x^2-7x+10$. 3

OR

A rectangle has vertices $A(-2, 3)$, $B(22, 3)$, $C(22, 35)$ and D. Find D and the length of diagonal AC. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Show that the points $A(1, 1)$, $B(5, 2)$, $C(7, 5)$ and $D(3, 4)$ taken in order form a parallelogram. 5

OR

The cost of x notebooks is a linear polynomial: a shop charges ₹45 for the bag plus ₹15 per notebook. Write the cost $C(x)$, find the cost of 4 notebooks and how many notebooks can be bought for ₹150. 5

27. Find the decimal expansion of $\frac{1}{13}$ by long division and state the length of the repeating block. 5

OR

Expand $(x+3y+1z)^2$. 5

END OF QUESTION PAPER — **Total: 50 Marks**

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